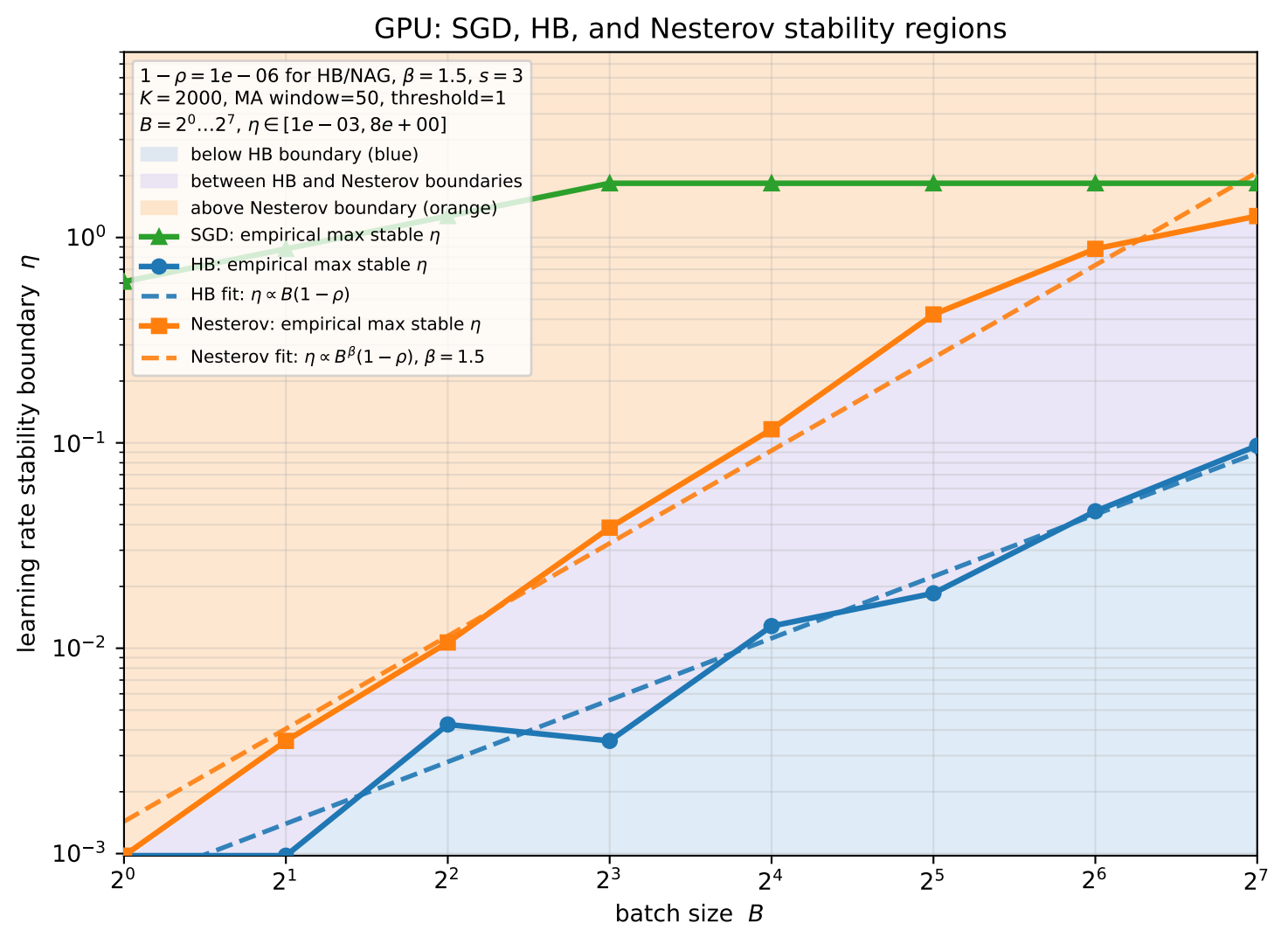
GPU: SGD, HB, and Nesterov stability regions
$1-\rho = 1e-06$ for HB/NAG, $\beta = 1.5$, $s = 3$
$K = 2000$, MA window=50, threshold=1
$B = 2^0 \ldots 2^7$, $\eta \in [1e-03, 8e+00]$
below HB boundary (blue)
between HB and Nesterov boundaries
above Nesterov boundary (orange)
SGD: empirical max stable η
HB: empirical max stable η
HB fit: $\eta \propto B(1-\rho)$
Nesterov: empirical max stable η
Nesterov fit: $\eta \propto B^{\beta}(1-\rho)$, $\beta = 1.5$
learning rate stability boundary η
10^{0}
10^{-1}
10^{-2}
10^{-3}
2^{0}
2^{1}
2^{2}
2^{3}
2^{4}
2^{5}
2^{6}
2^{7}
batch size B